# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, KOHAT

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

# MATHEMATICS

Past Paper 2023 • Paper Code: KT-X-23-I

Retyped faithfully from the original board paper — same questions, no changes.

## Section "A"

**Time Allowed:** 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** Code A

Note: Q.1 Attempt all questions of section-A by filling the correct option i.e. A, B, C or D on the OMR RESPONSE SHEET. Cutting, erasing and over writing will not be awarded any marks.

Qi. The solution of the equation $x^2 + 7x - 18 = 0$ are.....

(A) -2 or –9
(B) 2 or –9
(C) -2 or 9
(D) 2 or 9

Qii. If 'w' is the complex cube root of unity then $w^2$ = .....

(A) -1/w
(B) $w^3$
(C) 1/w
(D) 1

Qiii. Sum of the roots of $mx^2 - 5x + n = 0$ is .....

(A) 5/m
(B) -5/m
(C) n/m
(D) -n/m

Qiv. Direct variation between a and b is expressed as .....

(A) a = b
(B) a = 1/b
(C) $a \propto b$
(D) $a \propto 1/b$

Qv. Factors of $x^4 - x^2 - 6$ are .....

(A) $(x^2 + 3), (x^2 - 2)$
(B) $(x^2-3), (x^2 +2)$
(C) $(x^2 - 3), (x^2 - 2)$
(D) $(x^2 + 3), (x^2 + 2)$

Qvi. If A and B are any two sets then $\{x|x \in A \text{ and } x \notin B\}$ = .....

(A) B – A
(B) A'
(C) A – B
(D) [illegible in source scan]

Qvii. (sum of data)/(number of data) = .....

(A) Median (B) Mode

(C) Mean (D) Range

Qviii. In the set of data 5,7,7,5,3,7,2,8,2. The mode is .....

(A) 5 (B) 7

(C) 8 (D) 2

Qix. 45° = ..... radian.

(A) π (B) π/2

(C) π/3 (D) π/4

Qx. $\sin^2\theta + \cos^2\theta$ = ...

(A) 0 (B) tan θ

(C) 1 (D) –1

Qxi. If a,b,c are three sides of a triangle where 'c' is the largest side and $a^2 + b^2 = c^2$ then it is triangle.

(A) Right angle (B) acute angle

(C) obtuse angle (D) none of these

Qxii. Perpendicular from the center of a circle on a chord ..... it.

(A) intersects (B) Bisects

(C) trisects (D) divides

Qxiii. At any point on the circumference of a circle ..... tangents can be drawn.

(A) two (B) one and only one

(C) three (D) many

Qxiv. The angle whose vertex is on the circle and whose sides contain chords of the circle is ..... angle.

(A) inscribed (B) escribed

(C) right (D) acute

Qxv. If $2^x = 2$ then x = ...

(A) 0 (B) 1

(C) 2 (D) 4

## Section "B"

**Time Allowed:** 2 Hours 40 Minutes | **Maximum Marks:** 36

QQ.2. Attempt any NINE parts. All parts carry equal marks.

i Solve by factorization method $3(x^2 - 1) = 4(x + 1)$

ii Solve $3^{(2x)} - 10.3^x + 9 = 0$

iii For what value of k the roots of equation $x^2 + 5x + k = 0$ are imaginary.

iv The sum of the squares of three consecutive integers is 50. Find the integers.

v Find the fourth proportional of $a^3 - b^3$, $a + b$ and $a^2 + ab + b^2$

vi If y varies jointly as x and z, and y = 33 when x = 9 and z = 12, find y when x = 16 and z = 22

vii Resolve into partial fractions $1/((x+2)(x+1))$

viii If U = {1,2,3,4,5,6,7}, A = {2,5,6}, B = {1,2,3} then draw Venn diagrams for i. $A \cup B$ ii. $A \cap B$

ix If x = {1,2,3,4} and y= {5,6,7,8}, then write (i). one into function from x to y. (ii). one onto function from x to y.

x In a class of 13 students, the following test marks were obtained by the students. 10,12,12,14,9,18,9,13,16,9,17,16,14. Make a frequency table and find the Arithmetic mean.

xi Find the radius of the circle when the angle subtended with centre of the circle is $\pi/6$ radian and length of the arc is 2cm

xii Prove that $\cos^4\theta - \sin^4\theta = 2\cos^2\theta - 1$

## Section "C"

**Maximum Marks:** 24

Note: Note: Attempt any THREE of the following questions. All questions carry equal marks.

QQ.3. Prove that: one and only one circle can pass through three non-collinear points.

QQ.4. Prove that: the two tangents drawn to a circle from a point outside it, are equal in length.

QQ.5. Prove that: The angle in a semi-circle is a right angle.

QQ.6. Construct a triangle whose sides are 3cm, 4cm and 6cm. Draw its inscribed circle.